

HR DATA ANALYSIS PROJECT

1. Introduction

This project analyses an HR dataset to identify patterns relating to employee demographics, departments, income, job roles, job satisfaction, overtime, and attrition.

SQL was used to answer seven specified analytical questions, while Power BI was used for exploratory data analysis (EDA) and visualisation. The aim was to transform the employee data into meaningful insights that could support better understanding of the organisation's workforce.

2. Objectives

The project aimed to:

- Analyse the HR dataset using SQL.
- Examine employee income, departments, job roles, job satisfaction, overtime, and attrition.
- Use Power BI to explore patterns and relationships within the data.
- Present key findings through clear visualisations.
- Draw conclusions from the analysis.

3. Dataset And Data Preparation

The dataset contains 1,470 employee records and includes fields such as Employee Number, Gender, Department, Job Role, Monthly Income, Job Satisfaction, Over Time, and Attrition.

The HR data was extracted from the original Excel workbook and saved as a CSV file for SQL analysis. The CSV file was imported into SQL Server Management Studio (SSMS) as the HR_Data table.

The original Excel workbook was also imported into Power BI. The HR Data sheet was selected and the first row was set as the column headers before creating the visualisations.

4. Sql Analysis

Seven SQL queries were developed to answer the project questions. The analysis covered:

- Employees in Sales earning more than \$5,000 per month.
- The number of distinct job roles.

- Employee numbers and average monthly income by department.
- Male and female employees working overtime.
- Job roles with low average job satisfaction.
- Employee attrition and attrition rates by department.
- Employees earning above the company-wide average monthly income.

The SQL analysis provided the specific results required by the project and formed part of the basis for the Power BI exploration.

5. Power BI Analysis And Visualisation

Power BI was used to perform exploratory data analysis and present the results in a dashboard.

The dashboard contains two summary figures and six visualisations:

- Total Employees: 1,470
- Average Monthly Income: approximately \$6,500
- Employees by Department
- Average Monthly Income by Department
- Attrition by Department
- Employees by Job Role
- Average Job Satisfaction by Job Role
- Employees by Overtime and Attrition

The dashboard provides an overview of workforce distribution, income, job roles, employee satisfaction, overtime, and attrition.

6. Key Findings

Workforce Distribution

Research & Development has the largest number of employees (961), followed by Sales (446) and Human Resources (63).

Income

Sales has the highest average monthly income at \$6,859.17, while Research & Development has the lowest at \$6,261.25.

Job Roles

Sales Executive is the largest job role, with 326 employees, while Human Resources is the smallest, with 52 employees.

Job Satisfaction

Healthcare Representative has the highest average job satisfaction score (2.79), while Human Resources has the lowest (2.56).

Overtime and Attrition

A total of 127 employees who worked overtime left the organisation, compared with 110 employees who did not work overtime but left.

Among employees who worked overtime, approximately 30.5% left the organisation, compared with 10.4% among employees who did not work overtime. This indicates an association between overtime and higher attrition in the dataset, although it does not establish that overtime caused employees to leave.

7. Conclusion

The project demonstrated how SQL and Power BI can be used together to analyse HR data and communicate meaningful findings.

SQL provided specific answers to the required analytical questions, while Power BI allowed broader exploration of the dataset through visualisations. The analysis highlighted differences in workforce distribution, income, job roles, job satisfaction, and attrition, with overtime showing a notable association with employee attrition.

Overall, the project demonstrates how data analysis can turn raw employee records into useful information for understanding workforce patterns and identifying areas for further investigation.